Abstract

Many lightweight tactical UAVs have difficulty finding a compromise between directional stability and drag, while operating at Low Reynolds numbers close to Re ≈ 66,000. This can lead to flow separation, which can reduce aerodynamic performance. To address this problem, a computational framework was developed to investigate how different vertical tailfin geometries affect aerodynamic performance. The framework combines parametric geometry generation using CadQuery, Reynolds-averaged Navier-Stokes (RANS) simulations in OpenFOAM using the k-ω SST-LM model, Gaussian Process Regression (GPR) surrogate modeling, scalarized candidate selection, and NSGA-II multi-objective optimization. Near-wall mesh resolution was maintained with a mean y+ value of 1.6 (localized peaks around 30-70) as described in Section 2.2. The drag surrogate achieved an R^2 value of 0.991. The non-extrapolated reference design, run_25, exhibited a 36% increase in yaw-moment coefficient derivative ($Cn\beta$) and a 40.1% increase in side-force coefficient derivative ($CY\beta$) relative to the RQ-11 Raven vertical tailfin. These comparisons were evaluated at the same freestream velocity of 10 m/s, with moments referenced to the quarter-chord root location of each geometry. The improvements were accompanied by a 24.4% increase in drag coefficient (Cd). When normalized by drag, the yaw-moment and side-force derivatives increased by approximately 9% and 12%, respectively. A grid-convergence assessment was conducted using the Grid Convergence Index (GCI), with the fine-grid solution producing a GCI of 4.9%. Independent validation points were also evaluated using full RANS simulations, with drag prediction errors of +1.0% and −0.3%. Flow-field analysis at zero incidence indicated predominantly attached mean flow across the evaluated configurations, with no resolved separation bubbles identified in the retained wall-shear stress, near-wall velocity, and pressure-coefficient results.

1. Introduction

Small tactical unmanned aerial vehicles (UAVs) operate at relatively low Reynolds numbers, where viscous effects and boundary-layer transition can significantly influence aerodynamic performance. Their vertical tailfins must provide sufficient directional stability while minimizing parasitic drag, creating a design trade-off that is particularly important for lightweight airframes.

Tailfin aerodynamic performance is influenced by geometric parameters such as aspect ratio, taper ratio, and sweep angle. Although increasing fin area or modifying its spanwise distribution can improve directional response, these changes may also increase drag. Exploring this design space through individual computational fluid dynamics (CFD) simulations can be computationally expensive, motivating the use of automated simulation and surrogate-based optimization methods.

This study develops a computational framework for investigating low-Reynolds-number UAV tailfin geometries. The framework combines parametric CAD generation using CadQuery, Reynolds-averaged Navier–Stokes simulations in OpenFOAM using the k-ω SST-LM transition model, Gaussian Process Regression (GPR), and NSGA-II multi-objective optimization. A Latin Hypercube Design of

Experiments is used to sample the geometric design space, while surrogate models support candidate selection and aerodynamic trade-off analysis.

The framework is used to evaluate the relationship between drag and directional aerodynamic response, compare a non-extrapolated optimized configuration against an RQ-11 Raven reference fin, and examine the reliability of surrogate predictions. Mesh convergence, computational repeatability, and retained flow-field quantities are also assessed to identify the limitations of the numerical approach and the localized near-wall response metric used during optimization.

2. Methodology & Computational Pipeline

2.1 Parametric Geometry Generation (CadQuery)

All candidate vertical tailfin geometries were generated programmatically using a custom CadQuery parametric script. The geometric design space is governed by three primary non-dimensional shape parameters: quarter-chord sweep angle (Λ), aspect ratio (AR), and taper ratio (λ).

To maintain a consistent physical scale across the design space, the root chord (c_{root}) was fixed at 0.15 m. Tip chord and total span (b) were derived analytically according to:

$$C_{tip} = C_{root} \cdot \lambda$$

$$C_{mean} = \frac{C_{root} + C_{tip}}{2}$$

$$b = AR \cdot c_{mean}$$

C_{mean} represents the arithmetic mean chord. Spanwise sweep was implemented as a linear spatial offset applied to the tip profile. The resulting 3D volume was created as a straight ruled surface loft between the root and tip sections without spanwise twist or relative thickness variations.

Every candidate geometry in the optimization family utilizes the symmetric SD8020 airfoil profile, with approximately 7% thickness. To prevent high-aspect-ratio cell distortion and numerical divergence along the aft section during volume meshing, the sharp trailing edge from the raw coordinate file was blunted to a uniform 0.1% chord gap fraction across both root and tip sections.

2.1.1 Benchmark Comparison Framing

To benchmark performance against a representative tactical low-Re airframe, an RQ-11 Raven vertical tailfin geometry was constructed with the same CAD pipeline. The benchmark retains the stock Raven physical envelope: 100mm root chord, 60mm tip chord, 150 mm span, taper ratio of 0.6, sweep angle of 15°, and a planform area of 0.0120 m^2, utilizing an analytic NACA 0009 airfoil with 9% relative thickness.

By contrast, run_25 (case_25_b+000, Cd = 0.01085), serves as the non-extrapolated reference design, as it lies within the convex hull of the original 30-point DoE dataset. The surrogate-selected optimum, case_99, was identified through scalarized surrogate optimization. Although its design variables remain within the prescribed rectangular bounds, it lies outside the convex hull of the training dataset and is therefore classified as an extrapolative prediction. Its surrogate-predicted drag coefficient was $C_d = 0.0096977$, compared with a full-RANS confirmation value of $C_d = 0.0099824$.

The run_25 geometry features an SD8020 profile at 7% thickness and a larger planform area of 0.0195 m^2, versus the Raven's NACA 0009 at 9% thickness and 0.0120 m^2. To maintain consistency in the numerical setup between the two distinct geometries, both fins were evaluated using the same freestream velocity ($U\infty = 10$ m/s), identical snappyHexMesh boundary layer refinement rules, identical force reference parameters (A_{ref}, l_{ref}), and identical solver convergence criteria. Due to differences in characteristic length, the resulting Reynolds numbers were approximately 66.6k for run_25 and 44k for the Raven. Comparative performance and efficiency claims are evaluated using the yawing-moment and side-force coefficient derivatives normalized by the zero-sideslip drag coefficient ($C_{n\beta} / C_d$ and $C_{Y\beta} / C_d$).

2.2 CFD Solver Setup & Mesh Configuration

Numerical simulations were conducted using OpenFOAM v2606 running simpleFoam, a steady-state incompressible solver implementing the SIMPLE (Semi-Implicit Method for Pressure-Linked Equations) algorithm. Turbulence was modeled using the four-equation k-ω SST-LM transition model. To ensure solution convergence, a dual-criterion check was used: residuals had to decrease below 10^{-4} across all fields after running for at least 300 steps, and the force coefficients (C_Y, C_d, C_n) had to completely flatten out.

The domain was set from (-0.5, -0.5, -0.1) m to (1.5, 0.5, 0.5) m, making a total volume of 2.0 by 1.0 by 0.6 m. Relative to the 0.15 m root chord, this placed the inlet approximately 3.3c upstream of the fin and the outlet approximately 9c downstream. These dimensions were kept relatively compact to reduce computational cost while limiting blockage effects.

For boundary conditions, the inlet used a fixed velocity, the outlet was set to a fixed-value kinematic pressure outlet ($p = 0$ m^2/s^2) , and the sides and top used slip conditions. The bottom

plane at z = -0.1 m was set as a symmetry plane, leaving a 0.1m gap between the plane and the tailfin root at z = 0 so it didn't create an artificial wall junction. Sideslip angles were tested by rotating the incoming velocity vector. All aerodynamic coefficients were calculated using a reference area of 0.05 m² and a reference length of 0.149927 m. During the original simulations, moments for both geometries were calculated about a common absolute chordwise pivot at x = 0.0375 m. This location corresponds to the quarter-chord position of run_25, whose root chord is 0.15 m. However, the Raven has a 0.10 m root chord, placing its geometric quarter-chord at x = 0.025 m. Therefore, the Raven yawing moments were re-referenced from the original simulation pivot to its geometric quarter-chord location during post-processing before calculating the comparative yaw-moment derivative.

The meshes were generated from a blockMesh background grid (80 x 40 x 20, or 64k base cells) and subsequently refined using snappyHexMesh. This produced final cell counts between 200k and 420k for the design variations (for instance, the optimal design had 205,802 cells), with a hard cap set at 600k.

Near-wall regions were resolved using 12 inflated prism layers growing at a 1.2 expansion ratio from an initial first-cell height of 15μm, achieving around a 98% layer coverage. Regional wall-normal resolution yielded local y^+ means ranging between 0.7 and 3.1, while localized peaks of 30-70 occurred near stagnation regions and areas of high surface curvature. Overall, the mesh provided near-wall resolution across most attached boundary-layer regions, although localized increases in y^+ occurred near stagnation and high-curvature regions. A consistent near-wall meshing strategy was applied across the design family to minimize differences in wall-normal discretization between candidate geometries. All cases used the same initial prism-layer height, number of layers, and expansion ratio, with approximately 98% prism-layer coverage. This provides a consistent basis for comparing the localized near-wall response across geometries, although variations in local wall shear and surface curvature can still produce differences in y^+. Consequently, y^+ was treated as a comparative near-wall diagnostic rather than a direct measure of boundary-layer separation severity.

2.3 Design of Experiments and Data Collection

The design space was sampled using a 30-point Latin Hypercube over sweep, with sweep angles ranging from 15° and 35°, aspect ratio between 1.5 and 3.0, and taper between 0.4 and 0.8. Each design was built into a complete case directory from the frozen template by an orchestration script. The workflow consisted of a fresh CAD export at coarse tessellation, trimesh watertight repair, blockMesh, surfaceFeatureExtract, and snappyHexMesh -overwrite, followed by simpleFoam under the monitored convergence policy. (a minimum of 300 iterations and a maximum of 1,000 iterations, with termination permitted only when all field residuals fell below 10^{-4} across consecutive checks and drag and lift remained stable over a 50-iteration window).

Each case was executed within an exception handling routine,so any meshing failure or floating-point divergence was logged to an error file, marked FAILED in the master table, and the sweep continued without intervention. Force coefficients were recorded at every iteration using two function objects: a longitudinal forceCoeffs and a lateral forceCoeffsLateral object, whose lift and pitch channels read out side force (C_Y) and yaw moment (C_n), respectively. The yaw moments were initially calculated about the common absolute pivot at x = 0.0375 m, with the Raven results subsequently re-referenced to its geometric quarter-chord location during post-processing.

Flow-regime checks were performed during post-processing by mapping per-face y^+ values to local chord fractions and calculating statistics over the predefined 30–70% chord region. This interval was selected as a consistent mid-chord region for comparing localized near-wall behaviour across the candidate geometries, while excluding the immediate leading-edge and trailing-edge regions, where strong geometric and flow-gradient effects can complicate direct comparisons. The interval was therefore used as a fixed spatial sampling window rather than an assumption that the laminar separation bubble occupies the same chordwise location in every configuration. Near-wall flow behaviour was examined using wall-shear stress, streamwise velocity, turbulence kinetic energy, and intermittency fields retained from the SST-LM simulations. Wall-shear stress and near-wall streamwise velocity were used to assess the presence of local flow reversal, while intermittency and turbulence kinetic energy were examined to characterize the development of transition within the boundary layer. Intermittency was sampled at fixed wall-normal distances within the resolved near-wall region to avoid contamination from freestream values and coarse outer-layer cells. Pressure-coefficient distributions were also inspected for pressure plateaus and recovery patterns associated with separated flow. These diagnostics were used collectively to assess whether separation and reattachment were present, rather than inferring separation-bubble locations from a single proxy.

2.4 Surrogate Modeling and Optimization Setup

Four response targets (C_d, C_l, y^+ mean, and average y^+) were fitted using three model families: Gaussian Process Regression (GPR) with a Matérn kernel, gradient boosting, and quadratic polynomial response surfaces with ridge regularization. Models were compared using five-fold cross-validation, with performance assessed through R^2, out-of-fold RMSE and MAPE, and leave-one-out RMSE. The best-performing model for each response target was retained for subsequent optimization.

Candidate selection was performed using a scalarized surrogate objective:

$$J' = C_d + \alpha \cdot y^+_{30-70\%}$$

where $y^+_{LSB,\,mean}$ represents the mean y+ within the predefined 30–70% chord region, and α =0.005 is the weighting coefficient assigned to this localized boundary-layer proxy. A dense random search was conducted by evaluating 20,000 candidate geometries sampled within the observed minimum and maximum bounds of the design variables. The candidate minimizing the predicted scalarized objective was selected as the surrogate-based optimum. This procedure identified case_99, with predicted values of Cd = 0.0096977 and LSB-zone mean y+ = 0.69033.

The scalarized objective represents a computational proxy rather than the originally proposed aerodynamic objective, which incorporated side-force response normalized by drag and independently evaluated separation characteristics. Because the axial-flow dataset did not contain the sideslip simulations required to determine directional derivatives, these quantities could not be evaluated consistently across the complete design space. The mean y^+ within the predefined 30–70% chord region was therefore used as an auxiliary optimization target. This quantity characterizes localized near-wall response and is not a direct measure of boundary-layer separation, transition location, or separation-bubble length. Separation behaviour was assessed independently through wall-shear stress, near-wall velocity, intermittency, turbulence kinetic energy, and pressure-coefficient distributions.

Multi-objective optimization was conducted separately using NSGA-II, with drag coefficient (Cd) and LSB-zone mean y+ treated as independent objectives. The algorithm used a population size of 100 over 100 generations within the prescribed design-variable bounds. Unlike the scalarized random-search procedure, NSGA-II was used to characterize the non-dominated trade-off between the two surrogate-predicted responses rather than to select the single reference optimum. Candidate locations outside the convex hull of the original training dataset were explicitly identified as extrapolations, even when they remained within the rectangular design-variable bounds.

2.5 Verification of Mesh, Model and Repeatability

Grid independence was assessed on the surrogate-selected optimum geometry using three mesh configurations comprising 84,978, 205,802, and 448,308 cells. All configurations retained an identical blockMesh background grid of 80 x 40 x 20 cells, with surface refinement levels increased from (2, 3) to (3, 4) and (4, 5), respectively, using snappyHexMesh. The 12-layer prism configuration, including its 15 μm initial first-cell height and 1.2 expansion ratio, was held constant across all three meshes. The corresponding drag coefficients were

0.01128/0.00998/0.00951, monotonic with a fine-grid GCI of 4.9% and a Richardson-extrapolated value of 0.00913.

Turbulence-model sensitivity was assessed by comparing the four-equation k-ω SST-LM transition model against the two-equation k-ω SST model on two geometries. The turbulence model was the only configuration change between corresponding simulations. For case_99, the SST model predicted a drag coefficient of 0.01023, compared with 0.00998 using SST-LM, representing a 2.5% increase. For hi_07, the corresponding predictions were 0.00441 and 0.004316, respectively, representing a 2.2% increase. These comparisons indicate that the choice of turbulence model introduced approximately 2.1-2.5% variation in predicted drag across the two evaluated geometries.

Computational repeatability was assessed by independently regenerating and rerunning selected geometries through the complete computational pipeline, including CAD export, background mesh generation, surface refinement, and steady-state RANS simulation. For the run_25 geometry, two independently generated simulations produced drag coefficients of 0.01085055 and 0.01085045, with both meshes containing 230,378 cells. The relative difference between the results was approximately 9×10^{-6}, demonstrating highly consistent drag predictions across independently executed computational workflows.

3. Results

3.1 Pareto Frontier and Multi-Objective Design Space Topography

The NSGA-II multi-objective optimization mapped the non-dominated trade-off between zero-sideslip parasitic drag and the localized boundary-layer proxy, as shown in Figure 1. The feasible design space exhibits a distinct degenerate corner collapse, where candidate configurations cluster tightly toward lower drag values while demonstrating variable near-wall boundary-layer metrics across the sampled parameter space.

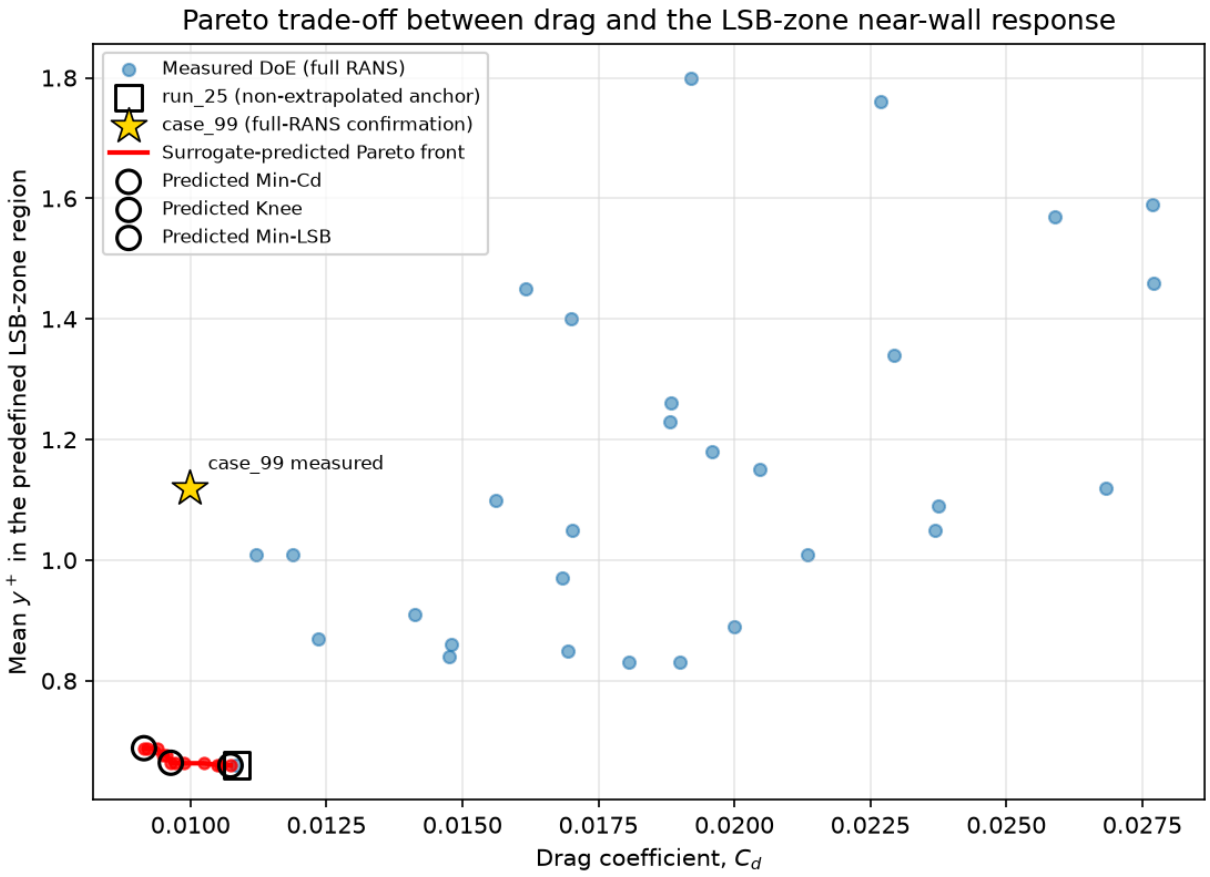


Figure 1: NSGA-II Pareto frontier mapping the trade-off between drag coefficient and the localized mean y^+ response within the predefined 30–70% chord region across the 30-point DoE dataset. The yellow star identifies case_99, the separately selected scalarized-surrogate optimum, with its full-RANS confirmation result shown for comparison.

3.1.1 Macro-Level Trade-Offs and Degenerate Corner Collapse

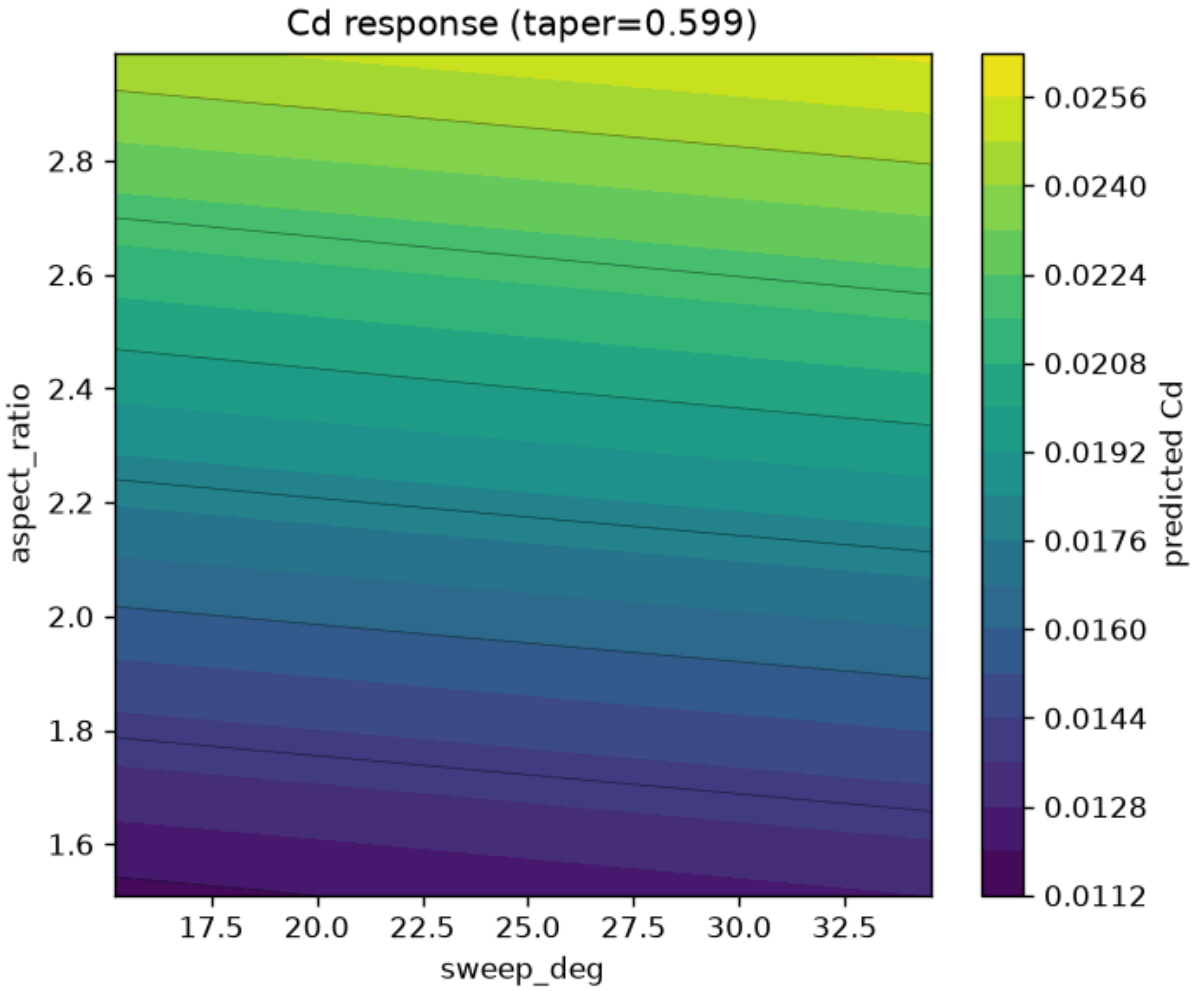


Figure 2a: Response surface contour illustrating the predicted drag coefficient (C_d) as a function of sweep angle and aspect ratio at a fixed taper ratio of 0.599.

The NSGA-II multi-objective optimization mapped the tradeoff between parasitic drag (C_d) and the mean y^+ within the predefined candidate separation region, used as a localized near-wall response proxy. As illustrated by the surrogate-driven Pareto analysis, the feasible design space exhibits a distinct degenerate corner collapse. Within the sampled design space, taper ratio exhibited the strongest first-order influence on the localized mean y^+ response.

Lower taper ratios were associated with lower predicted values of this near-wall diagnostic. This relationship describes the surrogate's predicted response within the sampled parameter space and should not be interpreted as evidence of reduced boundary-layer separation or delayed transition. As visualized in the response surface contours of Figure 2a, parasitic drag is heavily governed by the aspect ratio rather than the sweep angle at a fixed taper ratio; increasing the aspect ratio drives a monotonic escalation in C_d from approximately 0.011 to over 0.025, aligning with its dominant first-order sensitivity.

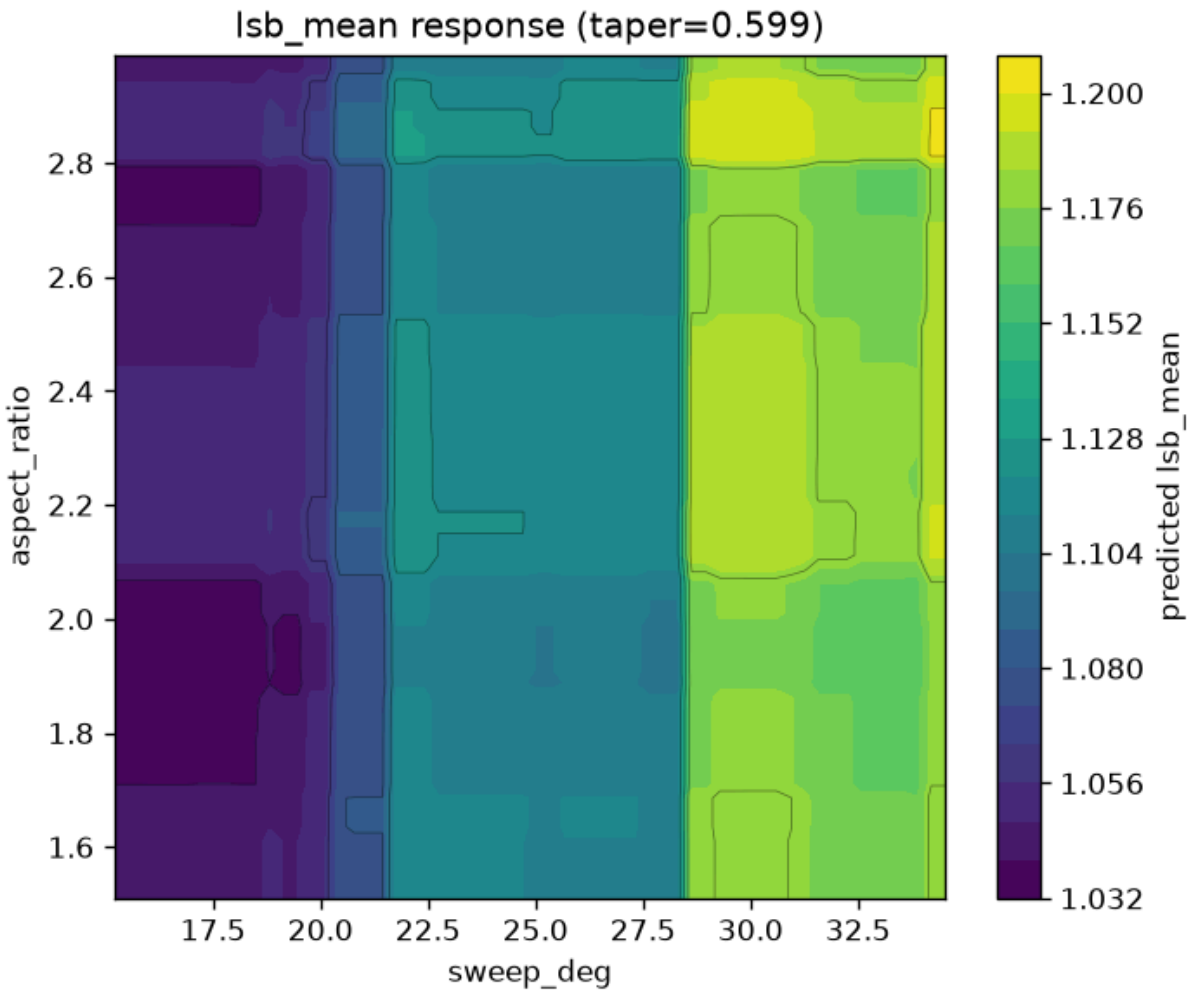


Figure 2b: Response surface contour illustrating the predicted LSB-zone mean y^+ proxy as a function of sweep angle and aspect ratio at a fixed taper ratio of 0.599.

In contrast, the corresponding LSB-zone mean y^+ response surface in Figure 2b exhibits distinct step-like stratification across sweep angles, reflecting the tree-based gradient boosting surrogate structure and highlighting how localized near-wall metrics respond non-linearly to planform sweep variations.

The resulting Pareto frontier identifies three representative design markers: the minimum-drag candidate, the knee point representing a compromise between drag and the localized boundary-layer proxy, and the candidate minimizing the LSB-zone mean y+ response. Because extracting directional derivatives required multi-angle velocity-vector rotations for selected geometries rather than the complete 30-point Design of Experiments matrix, the proxy-based optimization was used to identify candidate geometries for subsequent directional evaluation.

The surrogate-selected optimum, case_99, lies outside the convex hull of the original 30-point training dataset, although it remains within the prescribed design-variable bounds. Its full-RANS confirmation produced a drag coefficient of Cd = 0.0099824. However, its measured LSB-zone mean y+ of 1.12 differed from the surrogate-predicted value of 0.69033 (Figure 1).
This discrepancy demonstrates the reduced reliability of the localized near-wall proxy when evaluated outside the convex hull of the training dataset. In contrast, the drag prediction differed from the full-RANS result by approximately 2.85%, indicating better agreement for the integrated aerodynamic response. value should therefore be interpreted as an estimate of the localized near-wall diagnostic, not as a prediction of transition location or separation behaviour.

3.2 Aerodynamic Performance vs. RQ-11 Raven Baseline

To compare the aerodynamic performance of the non-extrapolated reference geometry, run_25, against the RQ-11 Raven baseline, both configurations were evaluated under the same freestream velocity of 10 m/s. Due to differences in their characteristic chord lengths, the resulting Reynolds numbers were approximately 66,600 for run_25 and 44,000 for the Raven.

Moments were evaluated about the quarter-chord root reference point of each geometry. For run_25, this corresponds to an absolute chordwise position of 0.0375 m, whereas the Raven's quarter-chord reference is located at 0.025 m. Re-referencing the Raven moment from the common 0.0375 m pivot used in the original comparison to its geometric quarter-chord position changes the calculated yaw-moment coefficient derivative through the associated side-force moment contribution. Accordingly, the reported derivatives are specific to the respective geometric reference points and should not be interpreted as a direct comparison of whole-aircraft directional stability. These reference locations define the isolated-fin comparison and are not estimates of the complete aircraft's centre of gravity. Using the respective quarter-chord references, run_25 exhibits an approximately 36% higher yaw-moment coefficient derivative ($Cn\beta$) than the Raven.

The increased directional response is consistent with the larger planform area and its spanwise distribution. Compared with the stock RQ-11 Raven geometry, run_25 has a larger planform area, increasing from 0.0120 m² for the Raven to 0.0195 m² for run_25. Its altered spanwise geometry also redistributes aerodynamic surface area farther outboard. Under sideslip, the resulting side-force distribution acts across a longer moment arm relative to the quarter-chord root reference, contributing to the increased yawing moment. The change from the stock 9%-thick NACA 0009 profile to the 7%-thick SD8020 profile introduces an additional difference in airfoil geometry; however, its independent contribution to the observed aerodynamic changes was not isolated. Because the two configurations differ in planform area, airfoil profile, and characteristic Reynolds number, the comparison represents an assessment of the two complete fin configurations under the specified freestream condition, rather than an isolated evaluation of planform geometry.

At the same freestream velocity, run_25 produces a 40.1% increase in the side-force coefficient derivative (CYβ) and an approximately 36% increase in the yaw-moment coefficient derivative (Cnβ), relative to the Raven when each is referenced to its own quarter-chord root. These increases are accompanied by a 24.4% increase in drag coefficient, from 0.00872 for the Raven to 0.01085 for run_25. Normalizing the derivatives by zero-sideslip drag gives an approximately 9% increase in Cnβ/Cd and a 12% increase in CYβ/Cd.

Thus, under the evaluated conditions, run_25 produces greater yaw-moment and side-force coefficient derivatives relative to its drag coefficient than the Raven baseline, despite its higher drag coefficient.

3.3 Flow-Field Physics and Boundary Layer Behavior

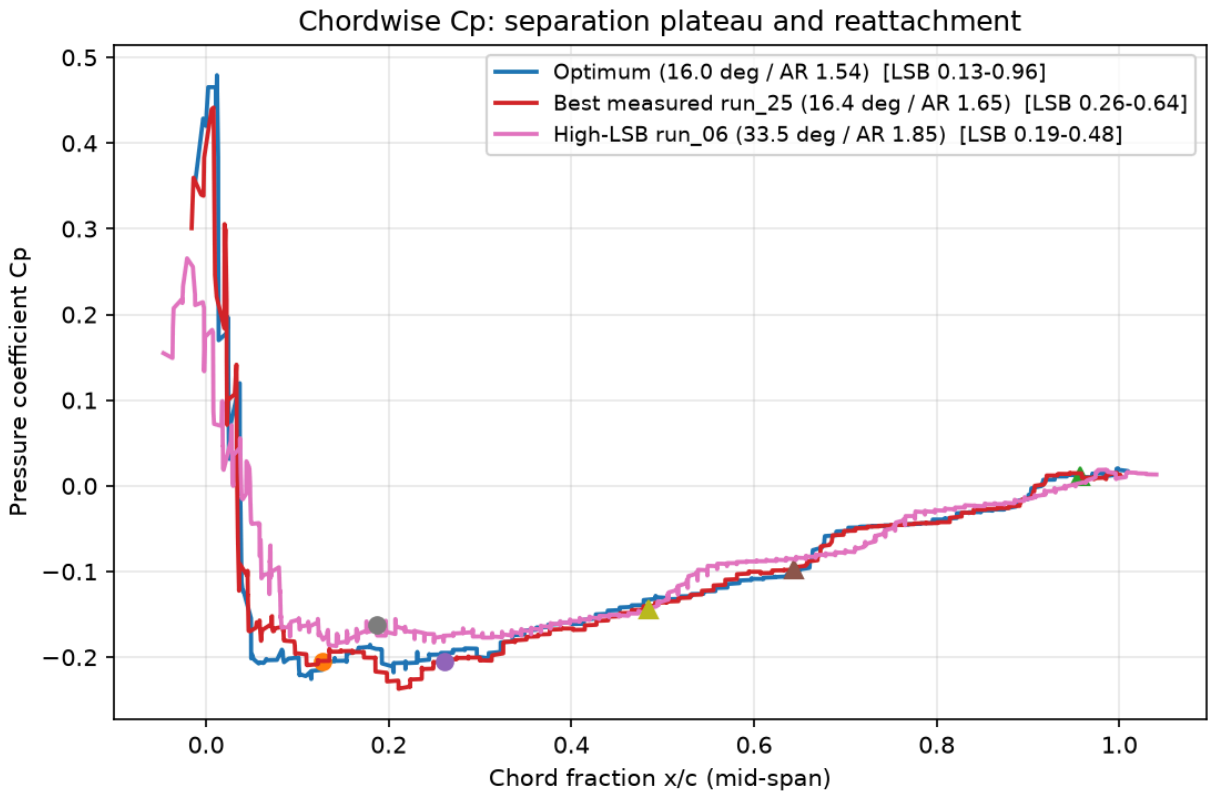


Figure 3: Chordwise pressure coefficient distributions at mid-span illustrating the separation plateau and reattachment characteristics for the surrogate optimum, run_25, and run_06 at zero incidence.

At the operational Reynolds number of approximately 66,600, the SST-LM simulations predicted attached mean flow across the mid-span regions examined for run_25 and case_99. The streamwise wall-shear stress remained negative across all sampled wall faces for both geometries, while near-wall streamwise velocity remained positive throughout the sampled

region. These results provide no evidence of mean-flow separation or a resolved separation bubble in either configuration. The RQ-11 Raven similarly exhibited predominantly attached flow, with only nine of 9,624 sampled wall faces showing small positive wall-shear values. These isolated values were negligible relative to the overall wall-shear distribution and were not interpreted as evidence of a resolved separation bubble.

The retained SST-LM intermittency and turbulence kinetic energy fields indicate a gradual increase in transition-related activity toward the trailing edge. Intermittency sampled at fixed near-wall distances remained low through most of the chord, with increases occurring primarily in the aft portion of the fin. This behaviour is consistent with late transition within an attached boundary layer rather than transition within a separated shear layer. Chordwise pressure-coefficient distributions likewise showed smooth suction and gradual pressure recovery, without a distinct separation plateau followed by abrupt recovery. Taken together, the wall-shear, near-wall velocity, intermittency, turbulence kinetic energy, and pressure-coefficient results support predominantly attached mean flow under the evaluated zero-incidence conditions.

The localized near-wall y^+ response used in the optimization should therefore be interpreted independently of separation behaviour. Although the surrogate-selected case_99 minimized the scalarized objective, the available flow-field evidence does not establish that it reduced separation or improved transition characteristics. All optimization evaluations were conducted at zero angle of attack. Sideslip polars were evaluated separately to quantify directional restoring moments, while angle-of-attack performance polars were outside the scope of the present study.

3.4 Surrogate Model Fidelity and Global Parameter Sensitivities

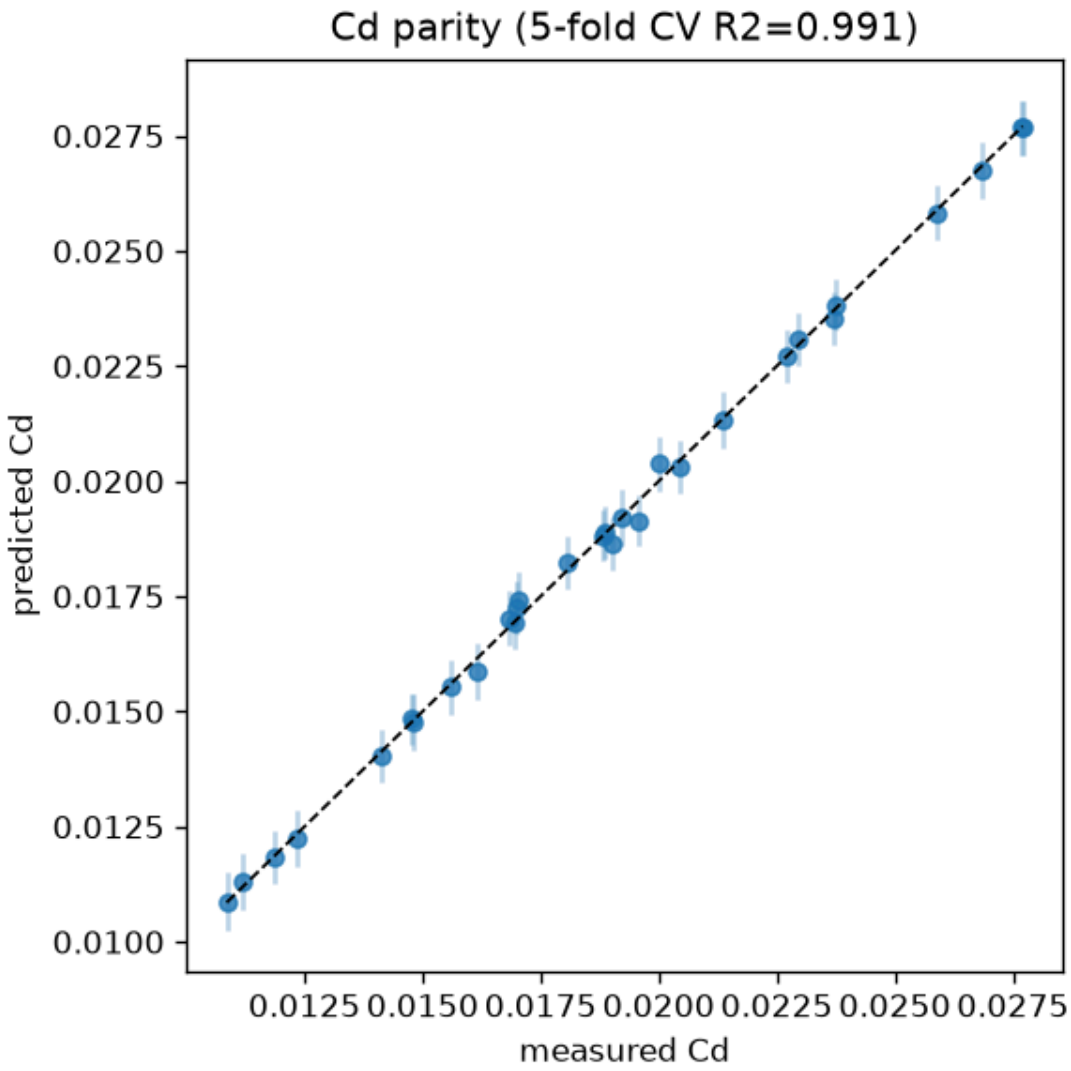


Figure 4: Parity plot of predicted versus measured drag coefficient C_d from 5-fold cross-validation, demonstrating high surrogate fidelity.

Gaussian Process Regression (GPR) with a Matern kernel yielded high predictive accuracy for integrated aerodynamic force coefficients: $R^2 = 0.991$ for drag (C_d) and $R^2 = 0.998$ for axial lift (C_l). As demonstrated by the tight clustering along the ideal diagonal in the 5-fold cross-validation parity plot of Figure 4, the drag surrogate achieves exceptional regression fidelity ($R^2 = 0.991$), with predictions aligning closely with full-RANS simulation values across the design space. Because axial lift for the symmetric vertical tailfin remains close to zero at zero incidence, the high R^2 value should be interpreted cautiously. In this case, the small absolute magnitude and limited variation of the lift coefficient make R^2 alone an insufficient measure of surrogate predictive performance.

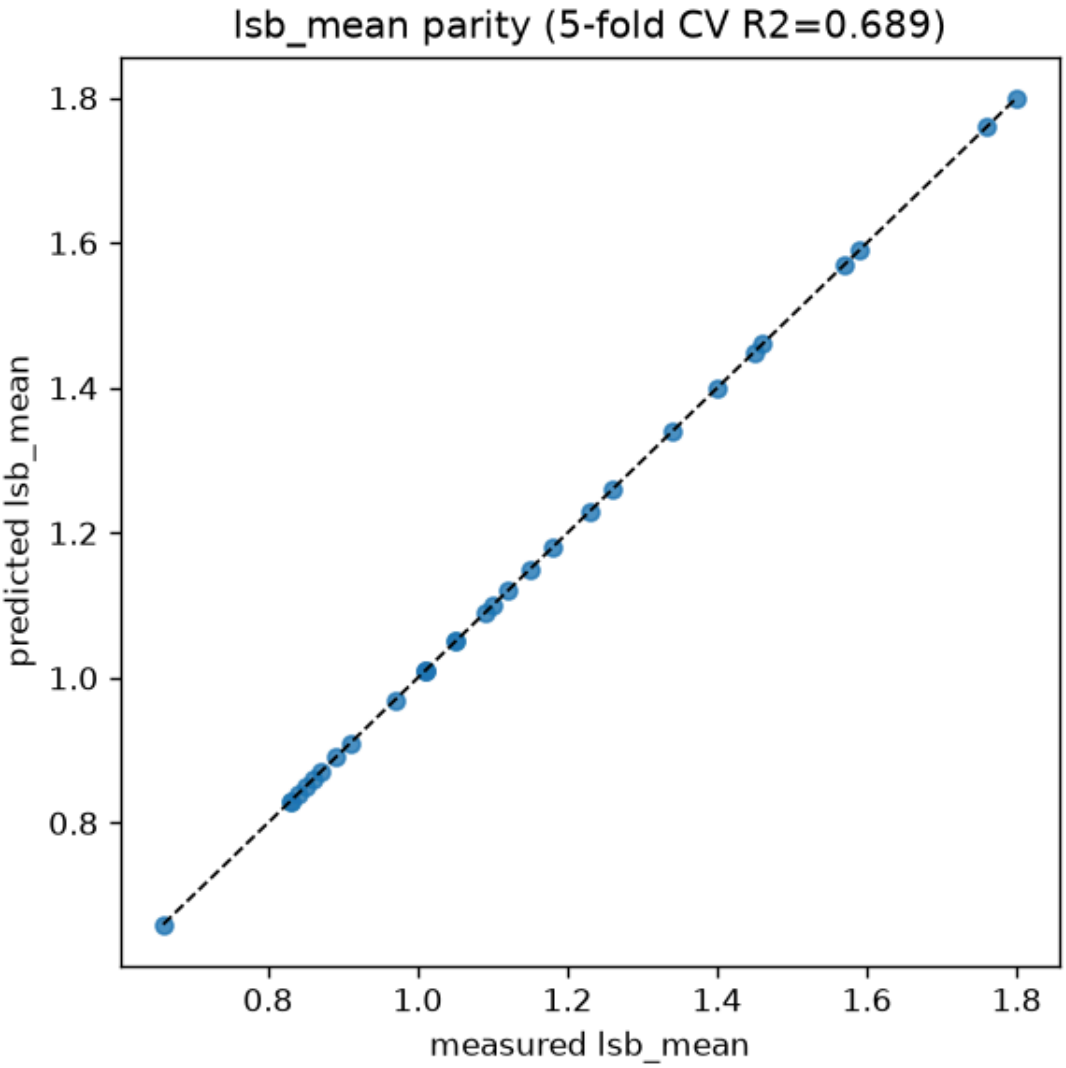


Figure 5: Parity plot of predicted versus measured LSB-zone mean y^+ proxy from 5-fold cross-validation, illustrating the gradient boosting surrogate performance.

In contrast, modeling the localized boundary layer separation proxy via gradient boosting yielded a reduced R^2 of 0.69. As shown in the 5-fold cross-validation parity plot of Figure 5, the gradient boosting surrogate achieves a clean internal fit, though its localized predictive variance remains higher than that observed for global integrated forces. The lower predictive accuracy for the LSB-zone response reflects the greater difficulty of modelling this localized flow-related quantity compared with the integrated aerodynamic coefficients.

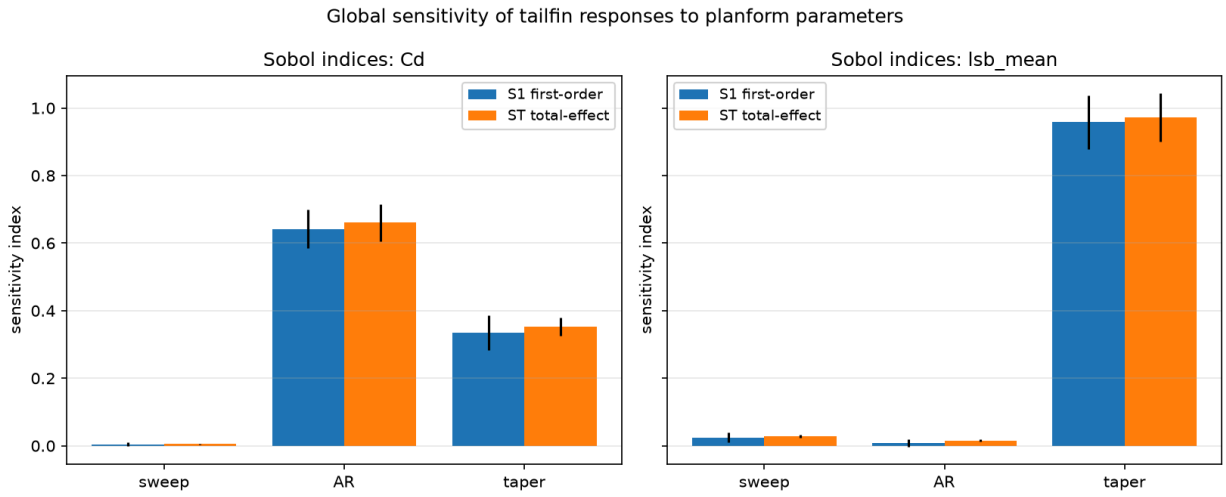


Figure 6: Global sensitivity decomposition showing first-order and total-effect Sobol indices for (a) drag coefficient and LSB-zone mean y^+ proxy across planform parameters.

Global variance-based sensitivity decomposition using Saltelli sampling established that aspect ratio had a first-order Sobol index of 0.642 for drag, while taper ratio exhibited a first-order index of 0.958 for the LSB-zone mean. Total-effect indices were similar to their corresponding first-order indices, suggesting that interactions between geometric parameters contributed relatively little to the overall response variance. As illustrated in the global sensitivity breakdown of Figure 6, these Sobol indices confirm that drag variance is driven primarily by aspect ratio, whereas localized boundary-layer proxy variance is almost exclusively dictated by taper ratio, with negligible interaction effects across the sampled design space. These results indicate that drag predictions were primarily sensitive to aspect ratio, whereas the localized near-wall response was predominantly influenced by taper ratio within the sampled design space.

4. Limitations and Future Work

4.1 Limitations of the Computational Framework

4.1.1 Transition Modelling at Low Re

Although the four-equation k-ω SST-LM model accounts for laminar-to-turbulent transition, its predictions remain dependent on empirical transition correlations and modelling assumptions. At the relatively low Reynolds numbers investigated in this study (approximately 44,000–66,600), these assumptions may introduce uncertainty into the predicted transition behaviour. The absence of experimental flow-field measurements also prevents direct validation of the predicted transition development. Consequently, the retained flow-field quantities provide a numerical assessment of near-wall behaviour rather than definitive experimental confirmation of transition characteristics.

4.1.2 Surrogate Proxy vs Direct Derivative Optimization

Computing full sideslip polars across the entire 30-point Design of Experiments matrix was computationally expensive. Consequently, the optimization loop relied on an auxiliary near-wall response proxy, while directional derivatives were evaluated separately for selected geometries. Although this approach provides a computationally efficient method for screening candidate configurations, the localized y+ response does not directly quantify boundary-layer separation, transition location, or directional aerodynamic performance.

Future optimization iterations could incorporate side-force and yaw-moment coefficient derivatives directly alongside drag, allowing candidate geometries to be evaluated against aerodynamic objectives more closely aligned with directional stability and efficiency. Additional flow-field diagnostics and experimental measurements would also help establish whether changes in the localized near-wall response correspond to meaningful aerodynamic improvements.

4.1.3 Rigid-Body Assumption and Aeroelastic Neglect

The computational framework treats all parametric geometries as rigid bodies and does not account for structural deformation under aerodynamic loading. Although the investigated aspect ratios remain within the prescribed range (AR = 1.5–3.0), the thin 7% SD8020 profile and slender planforms may exhibit structural deformation depending on material properties, construction, and operating conditions. Bending and torsional deformation could alter the effective fin geometry and aerodynamic response, particularly under increased aerodynamic loading or dynamic manoeuvres. These fluid-structure interaction effects were not captured by the present steady-state CFD framework.

4.2 Future Work

4.2.1 Transient Solvers and Dynamic Derivative Extraction

While steady-state RANS simulations (k-ω SST-LM) accurately resolve static stability derivatives ($C_{Y\beta}$, $C_{n\beta}$) at fixed angles of sideslip, they cannot capture time-dependent aerodynamic phenomena during dynamic maneuvering. To evaluate non-stationary flow behavior, future iterations of this framework will transition from steady-state models to unsteady RANS (URANS) or Scale-Adaptive Simulation (SAS) in OpenFOAM.

By implementing dynamic mesh motion, forced pitching and yawing oscillations can be prescribed according to:

$$\beta(t) = \beta_0 + \beta_A sin(\omega t)$$

where β_0 represents the mean sideslip angle, β_A is the oscillation amplitude, and ω is the reduced frequency.

This transient approach enables the direct extraction of dynamic damping and rotary derivatives, specifically the yaw damping derivative (C_{nr}) and roll-due-to-yaw derivative (C_{lr}). Furthermore, resolving time-accurate pressure fields will quantify phase lag between forced surface motion and transient side-force generation. Incorporating these dynamic derivatives will ultimately furnish full 6-DOF aerodynamic transfer functions for micro-UAV closed-loop flight control and autopilot tuning.

5. Conclusion

This study developed an automated computational framework for investigating and optimizing vertical tailfin geometries for lightweight UAVs operating at low Reynolds numbers. By combining parametric geometry generation, OpenFOAM simulations, surrogate modelling, and multi-objective optimization, the framework provided an efficient approach for exploring aerodynamic design trade-offs.

The drag surrogate achieved an R^2 of 0.991, while prediction of the localized near-wall y^+ response was less reliable. The separately selected candidate, case_99, demonstrated this limitation: its full-RANS drag prediction differed from the surrogate estimate by approximately 2.85%, while its localized mean y^+ prediction showed a larger discrepancy. This highlights the importance of full-CFD confirmation, particularly for candidates outside the convex hull of the training dataset.

Comparison against the RQ-11 Raven showed that run_25 produced 36% greater yaw-moment response and 40.1% greater side-force response, accompanied by a 24.4% increase in drag. When normalized by drag, these improvements were approximately 9% and 12%, respectively. Flow-field analysis indicated predominantly attached mean flow across the evaluated configurations, with no resolved separation bubbles identified at zero incidence. Consequently, the localized y^+ optimization metric should not be interpreted as a direct measure of separation or transition performance.

The numerical assessments, including a fine-grid GCI of approximately 4.9% and strong computational repeatability, support the consistency of the developed workflow. Nevertheless, experimental validation, improved near-wall resolution studies, and optimization using directly evaluated directional derivatives remain important areas for future work. Overall, the framework

demonstrates the potential of combining automated CFD and surrogate-based optimization for preliminary aerodynamic design of low-Reynolds-number UAV tailfins.